

Stress dependence of optimal blue excitation for nitrogen-vacancy optically detected magnetic resonance at 120 GPa

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Nitrogen-vacancy (NV) centers implanted in the culet of a diamond anvil cell are the reference local probe of magnetism in the megabar regime, where their usefulness is limited not by whether optically detected magnetic resonance (ODMR) survives compression but by how long a magnetic map takes to acquire. Once the anvil geometry is fixed, that time is controlled largely by one choice usually inherited from ambient-pressure practice: the excitation wavelength. We model the lock-in ODMR sensitivity $\eta \propto \Delta\nu/(C\sqrt{R})$ with all optical inputs anchored to the measured and calculated photophysics of the NV center up to 120 GPa, and with the Franck-Condon envelope evaluated at the temperature of the experiment. At 120 GPa the legacy single-mode baseline at stress anisotropy $\alpha = 0.95$ gives $\lambda_{\text{opt}} = 475.5$ nm, while the ground-state photoionization threshold imposes a hard blue boundary at 405 nm. Both conventional lines fall outside the 5% tolerance window 462–486 nm: 532 nm by a factor 2.5 because the absorption maximum has moved past it, and 405 nm by a factor 3.8 because it sits on the far blue flank with one-photon ionization switching on immediately beyond. Inside the window the steady-state charge state is nearly wavelength independent—excited-state ionization and recombination both scale with the absorption cross section and cancel from the NV[−] fraction—so the optimum coincides with the maximum of the photon-absorption rate $\lambda\sigma_{\text{abs}}$ and is insensitive to every phenomenological charge-transfer constant in the model. The commercial 473 nm line lies 2.5 nm away at a cost of 0.2% and is the practical recommendation within that baseline. A separate, hydrostatically anchored calculation uses the published Ho absorption kernel, whose optimum is approximately 441 nm, and adds only the phenomenological change with stress anisotropy. At fixed axial stress 120 GPa, its conditional optimum spans 478.9–493.4 nm at $\alpha = 0.56$ and 444.7–446.1 nm at $\alpha = 0.95$, returning to 440.65 nm at $\alpha = 1$. These intervals describe two stress normalizations, not confidence intervals; the anisotropic predictions remain unvalidated. The wavelength recommendation therefore requires a specified stress state and absorption model. Two further bounds apply to the legacy baseline. Below 73 GPa the same model returns green as the better choice, reproducing a published null comparison at 50 GPa; the resulting sign change is the prediction a single-pressure measurement cannot fake, and is accordingly the falsification test. And the wavelength is a low-intensity statement: the optimum moves into the blue as the transition saturates, so holding 473 nm at the upper end of the intensity range where existing high-pressure measurements operate costs a factor 1.5, comparable to the gain being claimed. Because the emission band shifts with the zero-phonon line as well, an inherited passband collects only 26% of the NV[−] light at 120 GPa; blue excitation is what makes it possible to move the detection cut-on with it, and the two gains compound to a factor $\simeq 4$. The validation checks twenty-two published high-pressure observations, reproduces twenty-one, and records one as unreproduced, spanning 0–150 GPa and four excitation wavelengths.

I. INTRODUCTION

The search for superconductivity in hydrogen-rich compounds, initiated by Ashcroft’s proposal that metallic hydrogen and hydrogen-dominant alloys should be high-temperature superconductors [1, 2], has produced the highest critical temperatures on record: 203 K in H₃S at 155 GPa [3] and 250–260 K in LaH₁₀ near 170 GPa [4, 5]. A second family of clathrate superhydrides operates at markedly lower pressure: CeH₉ is synthesized at 80–100 GPa and superconducts with T_c up to 115 K below one megabar [6, 7], and the substitutional alloy (La,Ce)H₉ reaches 148–178 K over 97–172 GPa [8]. These materials are the most attractive near-term targets precisely because their working pressures, clustered around 120 GPa, are compatible with a well-characterized diamond anvil cell (DAC) rather than with an extreme, low-yield one.

What has not kept pace is the *evidence*. A sample in a DAC at 120 GPa occupies a culet of a few tens of micrometers and a volume of order picoliters, so conventional bulk magnetometry cannot isolate its diamagnetic response from the anvil and gasket background [9]. Resistive signatures alone have repeatedly proved insufficient to settle claims of superconductivity in this field, which has made a direct, local, and spatially resolved measurement of magnetic flux expulsion the decisive experiment rather than a confirmatory one.

NV centers implanted a few tens of nanometers beneath the culet surface supply exactly this measurement. Because the sensor layer is fabricated *inside* the anvil, it sits micrometers from the sample and survives the same pressure the sample does; ODMR then reports the local magnetic field and the local stress tensor simultaneously [10–13]. This capability has been pushed from the 30 GPa scale to 130 GPa [14], to megabar pres-

tures [15, 16], and most recently to 240 GPa [17]. Its highest-profile application to date is the imaging of the Meissner effect and of flux trapping in CeH₉ at megabar pressures, which resolved the superconducting regions and showed that they are strongly inhomogeneous on the micron scale [18]; comparable measurements have since been reported for LaH₁₀ [19], for conventional superconductors under pressure [20], and for pressurized nickelates [21].

That last result identifies the bottleneck. The scientific content of these experiments is carried by *maps*—the spatial texture of the screening field, the geometry of trapped flux—and by their evolution through T_c and with pressure. A map is a raster of ODMR spectra, so the total measurement time scales as the number of pixels divided by the square of the magnetic sensitivity η_B [22]. In a regime where a single successful loading takes weeks to prepare and can be lost in one anvil failure, sensitivity is not a figure of merit but the currency in which spatial resolution, field-of-view, and the number of accessible (P, T) points are all purchased. The relevant question for NV sensing at 120 GPa is therefore not whether ODMR survives compression—recent photophysical work shows convincingly that it does [23, 24]—but how to make it as sensitive as possible there.

The answer is constrained by the fact that the NV center at 120 GPa is optically a different emitter from the one at ambient pressure, in three ways that a recent combined *ab initio* and experimental study has now quantified [23]. First, the NV[−] zero-phonon line (ZPL) blue-shifts by more than 0.4 eV, from 1.945 eV to $\simeq 2.35$ eV (529 nm). The 532 nm line, which at ambient pressure excites comfortably into the phonon sideband, therefore lands *at* the absorption onset at 120 GPa, where the cross section has collapsed. Excitation must move into the blue; this is a constraint, not a preference. Second, the electron-phonon coupling strengthens substantially—the absorption Huang-Rhys factor of the Jahn-Teller-inactive modes rises from 3.08 to 4.61 and the absorption Debye-Waller factor falls from 2.2% to 0.36%—so the absorption envelope both broadens and moves its weight to higher energy, placing its maximum near 474 nm. Third, the photoionization thresholds shift upward as well: IP(³A₂) rises from 2.68 eV to 3.06 eV and IP(³E) from 1.16 eV to 1.63 eV [23, 25]. Since 3.06 eV corresponds to 405 nm, the 405 nm diode line that is standard for charge-state manipulation sits exactly on the ground-state ionization edge at 120 GPa: it is the last wavelength before one-photon ionization out of ³A₂ switches on, and everything bluer converts NV[−] directly into NV⁰.

The three shifts do not act in the same direction, and that is what makes the problem nontrivial. Moving the excitation bluer follows the displaced absorption band and buys photon rate; moving it bluer also approaches the ground-state ionization edge, depletes the steady-state NV[−] fraction f_- , and dilutes the ODMR contrast with unpolarized NV⁰ fluorescence. The lock-in sensitivity, $\eta \propto \Delta\nu/(C\sqrt{R})$, weighs these against each other with

different powers—linearly in contrast, as a square root in photon rate—so where the balance falls is a quantitative question and not one that can be settled by inspecting the absorption spectrum. Reference [23] recognizes the issue and states the qualitative guideline that the excitation should be tuned to follow the ZPL shift, but stops short of the quantitative question that an experimenter has to answer when specifying a laser: *which* wavelength, to what tolerance, and whether that choice depends on the optical power—a question with direct consequences in a DAC, where the culet is small, the sample is metallic and absorbing, and the usable power is capped by heating rather than by the laser.

Here we answer that question. We construct a rate-equation model of the CW lock-in ODMR sensitivity in which the absorption cross section, the ionization thresholds, and the electron-phonon parameters are taken from the 120 GPa photophysics of Ref. [23], and only the charge-transfer rate constants and the NV⁰ background weight remain phenomenological. Minimizing η over the blue excitation window at 120 GPa yields an optimum at $\lambda_{\text{opt}} = 475.5$ nm in the legacy single-mode baseline with $\alpha = 0.95$. This is conditional on its envelope, rather than a universal blue-laser recommendation. Section IV A separately anchors the hydrostatic limit to the published Ho kernel and shows how a differential phenomenological stress correction changes that limit with α . The balance in the legacy baseline turns out to be resolved in a specific and testable way: within the blue window the two intensity-linear charge channels—excited-state ionization and recombination—both scale with σ_{abs} and cancel from f_- to better than 0.2%, so at fixed optical power the optimum is fixed by the photon-absorption rate and coincides with the maximum of $\lambda\sigma_{\text{abs}}$, while the ionization physics enters as a hard boundary at 405 nm rather than as a detuning of the optimum. We give the tolerance windows and the penalty of every commercial line, identify which photophysical anchors the answer is sensitive to, and—because a recommendation without its domain of validity is not usable—bound the answer on both sides: in pressure, where it reverses below 73 GPa, and in intensity, where it moves as the transition begins to saturate.

II. MODEL

A. Sensitivity functional

For continuous-wave, lock-in-detected ODMR the shot-noise-limited magnetic sensitivity is [22, 26, 27]

$$\eta_B \simeq \mathcal{P} \frac{h}{g\mu_B} \frac{\Delta\nu}{C\sqrt{R}}, \quad (1)$$

with $\Delta\nu$ the ODMR linewidth, C the resonance contrast, R the detected photon rate, and $\mathcal{P}$ a lineshape factor of order unity ($\mathcal{P} \simeq 0.7$ for a Lorentzian probed at maximum slope). Three features of Eq. (1) organize everything that follows. The contrast enters linearly and the

photon rate only as a square root, so a mechanism that halves C costs four times more than one that halves R ; the linewidth is independent of excitation wavelength at fixed power, so it cannot displace the optimum; and all three factors depend on the excitation wavelength only through the charge state and the absorption cross section.

Since we compare wavelengths rather than predict an absolute noise floor, we work throughout with the reduced quantity

$$\eta \equiv \frac{\Delta\nu}{C\sqrt{R}} \quad (2)$$

in arbitrary units, and quote results as ratios $\eta(\lambda)/\eta(\lambda_{\text{opt}})$. All overall scale factors—collection efficiency, NV density, integration bandwidth, $\mathcal{P}$ —cancel from such ratios. Converting to $\text{nT}/\sqrt{\text{Hz}}$ requires the calibration of Sec. VII and is deliberately not attempted here.

B. Anchored pressure-dependent inputs

The optical inputs are taken from the combined DFT and DAC study of Ref. [23] and carry no fitted freedom. The ZPL energy is represented by a saturating form,

$$E_{\text{ZPL}}(P) = 1.945 \text{ eV} + E_{\text{max}} \left[1 - e^{-P/P_0} \right], \quad (3)$$

whose two parameters are not free but are *derived* from two measured quantities: the initial slope $dE_{\text{ZPL}}/dP \rightarrow 5.75 \text{ meV GPa}^{-1}$ as $P \rightarrow 0$, measured to 60 GPa by Ref. [10], and the total shift $\Delta E_{\text{ZPL}}(120 \text{ GPa}) = 400 \text{ meV}$ reported by Ref. [23]. Requiring both gives $E_{\text{max}} = 0.570 \text{ eV}$ and $P_0 = 99.1 \text{ GPa}$. The slope has fallen to 1.0 meV GPa^{-1} by 120 GPa, which is why the linear extrapolation of Ref. [10] places the ZPL at 532 nm near 66 GPa while the saturating form places it at 112 GPa, in agreement with the observed slowing of the shift at high pressure [15, 28]. Parameterizing by the two measured quantities rather than by (E_{max}, P_0) matters for the uncertainty budget as well: it propagates measurement error instead of parameterization slack (Sec. IV D). Because $\Delta E_{\text{ZPL}}(120 \text{ GPa})$ is pinned, the initial slope has no influence on λ_{opt} at that pressure (Table V).

The absorption Huang-Rhys factor of the Jahn-Teller-inactive modes and the two photoionization thresholds are all reported to be nearly linear in P , and are interpolated between their end points,

$$S_{\text{abs}}(P) = 3.08 + 1.53 P/P_{120}, \quad (4)$$

$$\text{IP}(^3A_2)(P) = 2.68 \text{ eV} + 0.38 \text{ eV } P/P_{120}, \quad (5)$$

$$\text{IP}(^3E)(P) = 1.16 \text{ eV} + 0.47 \text{ eV } P/P_{120}, \quad (6)$$

with $P_{120} \equiv 120 \text{ GPa}$. The absorption Debye-Waller factor falls from 2.2% to 0.36% over the same range and

TABLE I. Anchored optical inputs, from Ref. [23]. These carry no fitted freedom; the wavelength equivalents are hc/E .

quantity	ambient	100 GPa	120 GPa
E_{ZPL} (eV)	1.945	2.297	2.345
equivalent (nm)	637	540	529
S_{abs}	3.08	4.36	4.61
DWF_{abs}	2.2%	0.5%	0.36%
K^2 (JT-active)	0.58	—	0.97
S_{em}	3.39	—	5.25
DWF_{em}	4.9%	—	< 1%
$\text{IP}(^3A_2)$ (eV)	2.68	3.00	3.06
equivalent (nm)	463	414	405
$\text{IP}(^3E)$ (eV)	1.16	1.55	1.63
equivalent (nm)	1069	799	761

is interpolated logarithmically. The effective phonon energy is held at $\hbar\omega = 65 \text{ meV}$ [29]. The underlying calculations were performed at fixed pressure through the diamond equation of state [30], so Eqs. (3)–(6) describe hydrostatic compression; the extension to a deviatorically stressed culet is Sec. VIII A. Table I collects the values at the pressures of interest.

Two features of Ref. [23] make these stronger anchors than a purely computational set would be. First, the absorption cross section is not only computed but checked against measurement at the two wavelengths that matter here: the integrated PL yield recorded under 532 nm and 457 nm excitation from ambient to 120 GPa traces two broad domes that follow the calculated σ_{abs} sampled at those same wavelengths. Since λ_{opt} turns out to be fixed by the photon-absorption rate $\lambda\sigma_{\text{abs}}(\lambda, P)$ (Sec. IV), the quantity our answer depends on is precisely the one that has been validated experimentally across the full pressure range. Second, because the excitation power and integration time were held constant in that comparison, the agreement also establishes that the NV radiative quantum efficiency is stable to 120 GPa, consistent with a computed radiative lifetime that varies by less than 1 ns over the same range. At fixed optical power the photon flux is I/E_γ , so we write $R \propto f_- I\sigma_{\text{abs}}/E_\gamma$ with no additional pressure-dependent radiative factor.

C. Absorption envelope at the temperature of the experiment

The absorption cross section is a single-effective-mode Franck-Condon envelope. It must be evaluated at finite temperature, and this is not a cosmetic refinement: at 120 GPa the 532 nm line lies 14 meV *below* the ZPL, so a $T = 0$ envelope—which vanishes identically below the ZPL—would assign green excitation a cross section coming entirely from the tail of the ZPL line, with a discontinuity where the ZPL sweeps past 532 nm near 113.7 GPa. That artifact predicts the death of green excitation at megabar pressure, in plain contradiction with room-temperature 532 nm cwODMR reported to

140 GPa [15].

We therefore use the standard single-mode multi-phonon lineshape at temperature T [31–33], with $\bar{n} = [e^{\hbar\omega/k_B T} - 1]^{-1}$,

$$L(p) = e^{-S(2\bar{n}+1)} \left(\frac{\bar{n}+1}{\bar{n}}\right)^{p/2} I_{|p|} \left(2S\sqrt{\bar{n}(\bar{n}+1)}\right), \quad (7)$$

continued to real $p = [E - E_{\text{ZPL}}]/\hbar\omega$, so that

$$\sigma_{\text{abs}}(E, P) \propto L\left(\frac{E - E_{\text{ZPL}}(P)}{\hbar\omega}\right) + \text{DWF}(P) g_{\text{ZPL}}(E), \quad (8)$$

normalized so that $\sigma_{\text{abs}}(532 \text{ nm}, 0) = 1$; g_{ZPL} is a narrow Gaussian of width 15 meV carrying the ZPL weight. Equation (7) is nonzero for $p < 0$ —the anti-Stokes, hot-band wing that keeps green excitation viable below the ZPL—and reduces exactly to the $T = 0$ Pekarian $e^{-S} S^p / \Gamma(p+1)$ for $p \geq 0$, and to zero for $p < 0$, as $T \rightarrow 0$. T is a measured experimental condition, not a fitted knob; we use 300 K throughout. Treating the phonon quantum number as continuous is appropriate at 120 GPa, where Ref. [23] reports that the sideband has lost its resolved replica structure entirely.

One consequence of that continuation matters for the results and is easy to get wrong. The *maximum* of the envelope is not the discrete Poisson mean $E_{\text{ZPL}} + S_{\text{abs}} \hbar\omega$. In the $T \rightarrow 0$ limit, where Eq. (7) reduces to $e^{-S} S^p / \Gamma(p+1)$, extremizing gives $\ln S_{\text{abs}} = \psi(p+1)$, whose root at $S_{\text{abs}} = 4.61$ is $p = 4.10$ rather than $p = S_{\text{abs}}$, placing the peak at 474.8 nm against 468.8 nm for the phonon-number mean. The $\simeq 6$ nm difference is comparable to the tolerances quoted in Sec. IV and must not be conflated with a physical detuning. At 300 K the thermal wings shift the maximum slightly further, and we locate it numerically rather than analytically: the finite- T envelope of Eq. (8) peaks at 474.0 nm at 120 GPa. All quoted maxima are numerical.

D. Charge-state dynamics and a cancellation

Under illumination NV^- is not stable: it can be ionized to NV^0 either by a one-photon process out of 3A_2 , which requires $E_\gamma > \text{IP}(^3A_2)$, or by a two-step process in which the center is first promoted to 3E and then ionized by a second photon [25, 34]. The two channels do not share a final state: ionization from 3A_2 leaves NV^0 in its 2E ground state, whereas ionization from 3E leaves it in the metastable 4A_2 state. Recombination proceeds through excitation of NV^0 , plus a dark channel. In steady state,

$$f_- = \frac{G_{\text{rec}}}{G_{\text{rec}} + G_{\text{ion}}}, \quad (9)$$

with

$$G_{\text{ion}} = \underbrace{a_{\text{gs}} [E_\gamma - \text{IP}(^3A_2)]_+}_{\text{one-photon, from } ^3A_2} + \underbrace{a_{\text{es}} \sigma_{\text{abs}}}_{\text{two-step, via } ^3E}, \quad (10)$$

$$G_{\text{rec}} = r_0 \sigma_{\text{abs}} + r_{\text{bg}}. \quad (11)$$

The bracket $[\cdot]_+$ denotes the positive part, so the first term switches on abruptly at the threshold and grows linearly with detuning above it. At 120 GPa this edge sits at 405.2 nm. The excited-state channel is open across the whole visible range at this pressure, since $\text{IP}(^3E) = 1.63 \text{ eV}$ corresponds to 761 nm.

Equations (9)–(11) contain the structural feature that dominates Sec. IV. Redward of the ionization edge the first term of Eq. (10) vanishes identically, and the two surviving rates are *both* proportional to σ_{abs} . Writing $\sigma \equiv \sigma_{\text{abs}}$,

$$f_- = \frac{r_0 \sigma + r_{\text{bg}}}{(r_0 + a_{\text{es}}) \sigma + r_{\text{bg}}} \xrightarrow{\sigma \gg r_{\text{bg}}/r_0} \frac{r_0}{r_0 + a_{\text{es}}}, \quad (12)$$

a constant. The wavelength dependence of the charge state survives only through the dark term r_{bg} , and therefore only where σ is small—that is, in the far wings, not near the optimum. This cancellation is a property of the model’s structure, not of a particular parameter choice, and it is what makes the low-intensity answer robust (Sec. IV E) and the intensity-dependent question interesting (Sec. V B).

E. Contrast, linewidth, and collection

The contrast is the product of an intersystem-crossing (ISC) prefactor and the dilution by unpolarized NV^0 fluorescence,

$$C = C_0(P) \frac{f_-}{f_- + w_0(1 - f_-)}, \quad (13)$$

where w_0 is the brightness of NV^0 relative to NV^- in the detection band. Contrast at high pressure is set by the ISC rather than by NV^0 dilution: the $^3E \rightarrow ^1A_1$ rate follows the phonon density of states resonant with the gap, which opens as the ZPL blue-shifts [35]. We therefore take $C_0(P) = C_{\text{amb}} \exp[-\Delta E_{\text{ZPL}}(P)/E_{\text{ISC}}]$, calibrated to the room-temperature 532 nm contrast of Ref. [15] (14% at ambient, $\simeq 3\%$ at 102 GPa, $\simeq 1.2\%$ at 138 GPa), giving $C_{\text{amb}} = 0.247$ and $E_{\text{ISC}} = 0.181 \text{ eV}$; the plateau observed above $\simeq 50$ GPa is produced by the saturation of ΔE_{ZPL} itself, so no separate floor is required. Making this prefactor explicit does not affect λ_{opt} —it is wavelength independent by construction, and the regression suite asserts that—but it makes the assumption visible and lets absolute contrast be quoted. The condition under which it would fail is examined in Sec. VIII B.

The linewidth carries a mild strain-broadening term,

$$\Delta\nu(P) = \Delta\nu_0 (1 + 0.15 P/100 \text{ GPa}), \quad (14)$$

which is wavelength independent and hence irrelevant to λ_{opt} at fixed power, but which matters for absolute sensitivity and for the power-broadened case.

Finally, $R \propto f_- I \sigma_{\text{abs}}/E_\gamma$ counts *absorbed* photons, whereas a confocal setup records the emission that survives a long-pass filter. The emission band is the mirror

image of the absorption envelope, built on the emission Huang-Rhys factor $S_{\text{em}} = 3.39 \rightarrow 5.25$ and $\text{DWF}_{\text{em}} = 4.9\% \rightarrow < 1\%$, and it blue-shifts with the ZPL. We therefore include a collection factor $\eta_{\text{col}}(P)$, the fraction of NV^- emission inside the passband. Emission leaves the relaxed 3E whatever vibronic level was excited (Kasha's rule), so η_{col} is wavelength independent and cannot move λ_{opt} ; it is nonetheless the missing piece in the pressure dependence of the detected count rate (Secs. IV G and VIII C).

F. Intensity-explicit extension

Everything above is linear in the excitation intensity I , so $\eta(\lambda)$ is almost independent of absolute power: scaling I only trades against the single intensity-independent term r_{bg} . That is a limitation of the model, not a statement about the physics, and it means the question of whether λ_{opt} depends on laser power cannot be posed within Sec. II D. We therefore introduce a dimensionless intensity u and restore the nonlinearities that CW-ODMR is known to have.

Fixing the intensity unit. The 3E population saturates,

$$n_E(\lambda, P, u) = \frac{q_\lambda \sigma_{\text{abs}} u}{q_\lambda \sigma_{\text{abs}} u + \Gamma_d}, \quad (15)$$

where $q_\lambda = E_{532}/E_\gamma = \lambda/532 \text{ nm}$ converts equal optical power to relative photon flux. Equation (15) is meaningful only once Γ_d and σ_{abs} share a scale, and σ_{abs} is normalized to green absorption at ambient pressure—an arbitrary unit. We therefore *define* u against a stated saturation condition: $u = 1$ is the intensity that half-saturates the NV^- transition at λ_{opt} at the reference pressure, i.e. $\Gamma_d = s_d q_{\lambda_{\text{opt}}} \sigma_{\text{abs}}(\lambda_{\text{opt}}(P_{\text{ref}}), P_{\text{ref}})$ with s_d a dimensionless residual knob equal to unity by definition. The low-intensity limit then restores $R \propto f_- q_\lambda \sigma_{\text{abs}}$ and the fixed-power model, which we verify as a regression test. An identical construction with its own decay scale Γ_{d0} defines n_{E^0} for NV^0 .

Two-photon ionization. The 3E channel is built from the excited-state population rather than postulated linear,

$$G_{\text{ion}}^{\text{ES}} = a'_{\text{es}} q_\lambda \sigma_{\text{abs}} u n_E, \quad (16)$$

which is quadratic in u at low intensity and crosses over to linear once the transition saturates. The one-photon ground-state channel remains linear, $G_{\text{ion}}^{\text{GS}} = a_{\text{gs}}[E_\gamma - \text{IP}({}^3A_2)]_+ u$.

Recombination through the NV^0 envelope. Recombination proceeds via excitation of NV^0 , which has its own Franck-Condon envelope evaluated at the same temperature. The NV^0 ZPL lies at 575 nm (2.156 eV), 0.21 eV above the NV^- ZPL [36], so that envelope is displaced to the blue—at 120 GPa it peaks near 439 nm against 474 nm for NV^- . The sign matters: it determines which way λ_{opt} moves with power (Sec. V B).

$$G_{\text{rec}} = r_0 \sigma_{\text{NV}^0}(\lambda, P) u n_{E^0} + r_{\text{bg}}. \quad (17)$$

TABLE II. Model inputs. Upper block: anchored to measurement, no fitted freedom. Middle block: phenomenological charge-transfer constants. Lower block: additional constants of the intensity-explicit extension. The last column is the range used in the Monte Carlo bands; entries marked “—” are held fixed.

symbol	meaning	value	MC range
$\Delta E_{\text{ZPL}}(120)$	measured ZPL shift	400 meV	$\pm 20 \text{ meV}$
$dE_{\text{ZPL}}/dP _0$	measured initial slope	$5.75 \text{ meV GPa}^{-1}$	± 0.25
∂S_{abs}	HR slope	$1.53/P_{120}$	$\pm 15\%$
$\hbar\omega$	effective-mode model	65 meV	$\pm 15\%$
σ_{ZPL}	ZPL line width	15 meV	$0.7\text{--}1.4\times$
T	temperature	300 K	—
α	stress anisotropy	0.95	—
a_{gs}	one-photon ionization	6.0	$0.5\text{--}1.6\times$
a_{es}	two-step ionization	0.9	—
r_0	recombination	1.2	$0.7\text{--}1.3\times$
r_{bg}	dark recombination	0.15	$0.6\text{--}1.5\times$
w_0	NV^0 brightness	1.0	—
$C_{\text{amb}}, E_{\text{ISC}}$	ISC prefactor	0.247, 0.181 eV	—
s_d	NV^- half-saturation	1.0	fixed by definition
s_{d0}	relative NV^0 saturation	1.0	$0.6\text{--}1.6\times$
a_{es}	two-photon ionization	0.9	$0.5\text{--}1.8\times$
Γ_c	power broadening	0.6	$0.5\text{--}1.8\times$
Γ_{sat}	contrast saturation	1.2	$0.5\text{--}1.8\times$

This is the step that breaks the cancellation of Eq. (12): once ionization scales with σ_{abs} and recombination with σ_{NV^0} , the two no longer divide out and f_- acquires a genuine wavelength dependence.

Saturation and power broadening. The detected rate becomes $R = f_- n_E$, which saturates, while the contrast is diluted and the line broadened by the optical pumping rate Γ_p in the standard CW-ODMR way [26],

$$C \rightarrow \frac{C}{1 + \Gamma_p/\Gamma_{\text{sat}}}, \quad \Delta\nu(u) = \Delta\nu(P) \sqrt{1 + \Gamma_p/\Gamma_c}. \quad (18)$$

The competition is then explicit: raising u increases R sublinearly while ionization grows superlinearly and the line broadens, so an interior optimum in u exists and λ_{opt} may move with it.

G. What is anchored and what is not

Table II separates the two classes of input. The distinction is essential to reading the results: conclusions that depend only on the upper block inherit the uncertainty of Ref. [23], whereas conclusions that depend on the lower blocks are provisional until the calibration of Sec. VII is done. As Sec. IV E shows, the headline result depends only on the upper block.

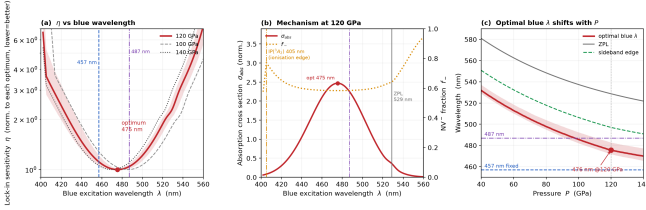


FIG. 1. (a) Reduced lock-in sensitivity $\eta(\lambda)$ at 120 GPa (normalized to its own optimum) with the Monte Carlo 16–84% band, compared with 100 GPa and 140 GPa. (b) Mechanism: the absorption cross section σ_{abs} and the steady-state NV^- fraction f_- , with the ZPL and the ground-state ionization edge marked. Note that f_- is flat across the entire region where σ_{abs} is appreciable. (c) The optimal blue wavelength as a function of pressure, tracking the ZPL and the sideband edge.

III. NUMERICAL METHODS

Sensitivity curves are evaluated on a wavelength grid spanning 402–560 nm. The lower bound is set by the ground-state ionization edge at 120 GPa and the upper by the point at which the cross section has fallen below 10^{-2} of its peak. A grid spacing of 0.05 nm is used for optimum extraction and 0.2 nm for two-dimensional maps. The location of the minimum is refined below the grid spacing by fitting a parabola to the three points bracketing the discrete `argmin`; this changes λ_{opt} by less than 0.02 nm at 0.05 nm spacing and is retained only so that derivatives with respect to model parameters are not quantized by the grid.

Uncertainty bands are Monte Carlo: each of n draws builds a complete model with every parameter marked randomizable in Table II sampled independently, the quantity of interest is evaluated, and the 16th and 84th percentiles are reported. We use $n = 250$ for curve bands, $n = 400$ for the distribution of λ_{opt} itself, and $n = 80$ for the two-dimensional power maps, where each draw costs a full-grid evaluation. The bands are reported as percentiles rather than standard deviations because the distribution of λ_{opt} is visibly asymmetric.

Anchor sensitivities are computed one at a time: each parameter is displaced to the edges of its range with all others held at their nominal values, and the resulting $\Delta\lambda_{\text{opt}}$ is tabulated. This is a local analysis and does not capture parameter correlations, which is acceptable here because the result (Sec. IV E) is that all but three sensitivities vanish identically.

All calculations run in a few minutes on a laptop. The code, the exact commands that regenerate each figure, and the regression tests that pin the quoted numbers are listed in Appendix A.

TABLE III. Conditional optimum at fixed $\sigma_{zz} = 120$ GPa and $T = 300$ K, from Eq. (20). The endpoints span the equal-axial-stress and equal-compression normalizations. The anisotropic predictions are unvalidated; the hydrostatic anchor is common to both.

α	λ_{corr} (nm)
0.56	478.9–493.4
0.60	475.2–488.1
0.70	466.1–475.4
0.95	444.7–446.1
1.00	440.7

IV. THE OPTIMUM AT LOW INTENSITY

A. Stress anisotropy changes the conditional optimum

We use compressive stress magnitudes and the axisymmetric convention

$$\alpha = \frac{\sigma_{xx}}{\sigma_{zz}} = \frac{\sigma_{yy}}{\sigma_{zz}}, \quad \bar{\sigma} = \frac{1+2\alpha}{3}\sigma_{zz}, \quad q = (1-\alpha)\sigma_{zz}. \quad (19)$$

Thus changing α at fixed $\sigma_{zz} = 120$ GPa changes both mean compression and deviatoric stress. Hydrostatic compression has $\alpha = 1$. The micropillar value $\alpha \simeq 0.95$ measured by Ref. [14] characterizes that validation geometry; it is not the stress assigned to the hydrostatic absorption calculation of Ref. [23].

The reconstructed hydrostatic Ho kernel supplies the interpolant optimum $\lambda_{\text{Ho}} = 440.65$ nm (reported as approximately 441 nm). To explore anisotropy without replacing that anchor by the single-mode envelope, we retain only the latter model’s differential shift, at the same axial stress and $T = 300$ K:

$$\lambda_{\text{corr}}(\alpha; k) = \lambda_{\text{Ho}} + \lambda_{\text{FC}}(\alpha; k) - \lambda_{\text{FC}}(1; k). \quad (20)$$

Here λ_{FC} is the phenomenological Franck–Condon model optimum, and its stress shift uses a factor $(1+2\alpha)/3 + k(1-\alpha)$. Equation (20) is anchored to the Ho value at $\alpha = 1$ for either normalization. It is a conditional hypothesis, not an anisotropic absorption spectrum measured or calculated by Ho et al.

The two ZPL slopes reported by Ref. [14], -434 and -769 meV/(cm³ mol⁻¹), constrain the sign of the deviatoric correction but leave its normalization unresolved. Equal compressed volume (equal mean compression) gives $k = -0.734$; equal axial stress gives $k = -0.39211$. Table III carries both readings. The range is an interval over model normalizations, not a statistical confidence interval.

Lowering α shifts this conditional blue optimum toward longer wavelengths. In particular, a 473 nm source cannot be selected for every geometry from the legacy 475.5 nm result. The static hydrostatic-kernel alternative, which holds 440.65 nm fixed for all α , is also untested away from hydrostatic stress. An absorption or

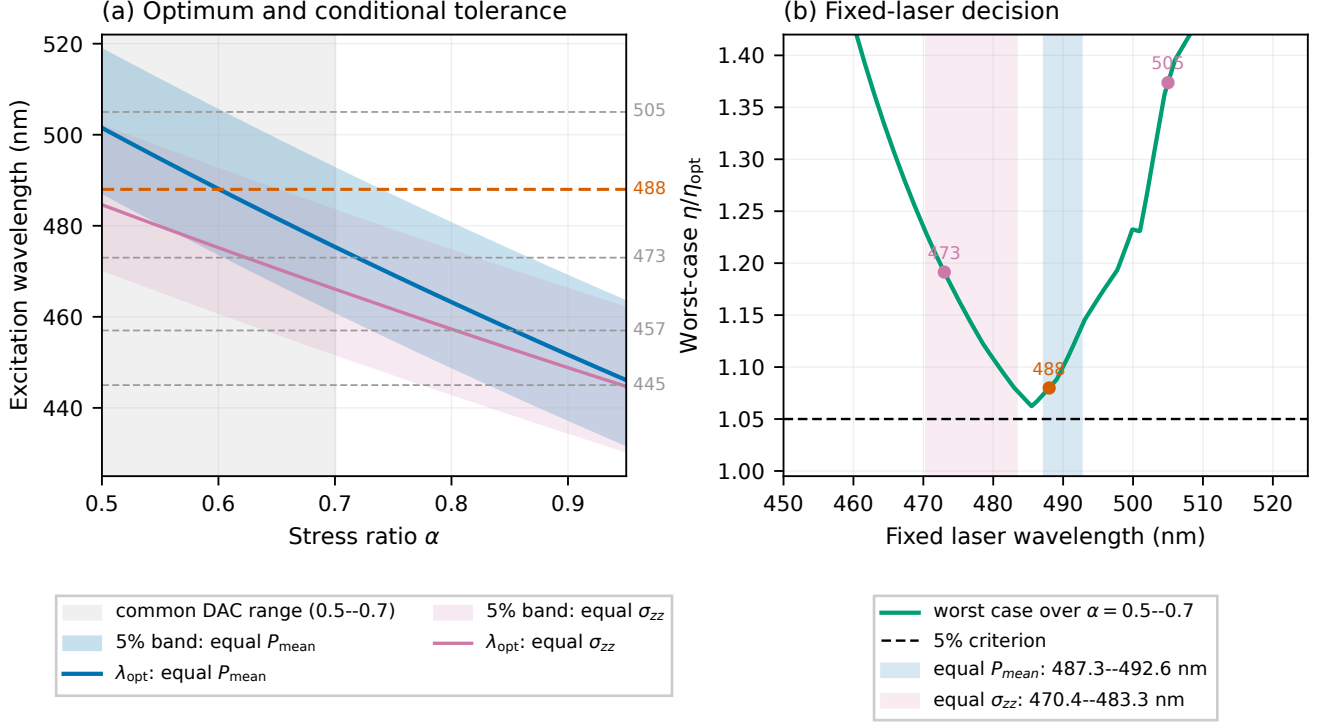


FIG. 2. Dependence on α at fixed axial stress 120 GPa: the differential phenomenological correction is anchored to the hydrostatic H_0 optimum of 440.65 nm. The two normalizations are $k = -0.734$ and $k = -0.39211$; their spread is not a confidence interval. The 5% tolerance bands use a rigid shift of the hydrostatic sensitivity shape. The stress coefficient is unknown, and these anisotropic predictions and shape shifts are unvalidated.

excitation sweep at known stress anisotropy is required to distinguish these hypotheses. The following subsections retain the legacy baseline's numerical results for comparison and reproducibility; its tolerance windows, pressure crossovers, and intensity shifts do not transfer automatically to the hydrostatically anchored correction.

B. Legacy baseline: result and mechanism

In the legacy $\alpha = 0.95$ single-mode baseline, at 120 GPa the anchored inputs give $E_{ZPL} = 2.345$ eV (528.7 nm), $S_{abs} = 4.61$, and $IP(^3A_2) = 3.060$ eV (405.2 nm). Minimizing Eq. (1) over λ yields

$$\lambda_{opt}(120 \text{ GPa}) = 475.5 \text{ nm}, \quad (21)$$

whereas the absorption cross section alone peaks at 474.0 nm. The 1.5 nm displacement is the $1/E_\gamma \propto \lambda$ photon-flux weighting required when wavelengths are compared at equal optical power rather than at equal photon flux.

That near-coincidence is a result rather than a triviality, and it identifies which physics sets the answer. Across the whole blue window the steady-state charge state is essentially constant: f_- varies from 0.5822 to 0.5833 over the 5% tolerance window, a spread of 0.2% [Fig. 1(b)].

This is the cancellation of Eq. (12) in action. With f_- constant, both C and the f_- factor in R are constant, and $\eta \propto (\lambda\sigma_{abs})^{-1/2}$; the minimum of η is then the maximum photon-absorption rate at fixed optical power. We verified that this is not an accident of the nominal parameters: setting $a_{gs} = 0$, raising r_{bg} from 0.15 to 1.0, or varying E_{max} and S_{abs} all leave λ_{opt} equal to the $\lambda\sigma_{abs}$ maximum to within 0.05 nm.

Ground-state ionization therefore does not displace the optimum. Its role is different and more absolute: it forbids the region blueward of 405.2 nm, where the positive-part term of Eq. (10) grows linearly with detuning and drives f_- down without bound.

This flatness is reached independently in Ref. [23], by a different argument. Plotting the ionization thresholds relative to the ZPL, that work observes that $IP(^3A_2)$ tracks the ZPL closely while the deformation potential of $IP(^3E)$ is only somewhat smaller, and concludes that if the excitation wavelength is adjusted to follow the pressure-induced shift of the ZPL, the charge-state stability of NV^- should remain comparable to ambient conditions. Our f_- result is the quantitative form of that statement, and the agreement indicates that the cancellation reflects the photophysics rather than an artifact of our parameterization.

TABLE IV. Sensitivity penalty $\eta(\lambda)/\eta(\lambda_{\text{opt}})$ of commercial laser lines at 120 GPa and room temperature (lower is better; 1.000 is optimal), with the cross section relative to its maximum.

λ (nm)	η/η_{opt}	$\sigma_{\text{abs}}/\sigma_{\text{abs}}^{\text{max}}$	comment
405	3.76	0.038	blue flank; at the IP(3A_2) edge
450	1.23	0.679	outside the 10% window
457	1.11	0.824	inside the 10% window
473	1.002	0.999	operationally optimal
475	1.000	0.999	nearest integer to the model optimum
488	1.05	0.874	inside the 10% window
505	1.33	0.509	outside
532	2.53	0.096	at the absorption onset

C. Commercial laser lines and tolerance

Table IV lists the penalty incurred by the commercially available laser lines. The two conventional choices fail for opposite reasons, and in both cases the cause is the collapse of the cross section rather than a charge-state effect. The 532 nm line, still the default in most NV setups, is degraded by a factor 2.5 because at 120 GPa it excites 3 nm to the red of the ZPL, where σ_{abs} has fallen to 26% of its ambient-pressure green value—a factor 10 below σ_{abs} at the optimum. The 405 nm line is degraded by the larger factor 3.8, because it lies far out on the blue flank, where $\sigma_{\text{abs}}/\sigma_{\text{abs}}^{\text{max}} = 0.038$; there the ground-state ionization term contributes only 8% of the total ionization rate, since 405 nm sits barely above threshold. Its real significance is the absence of room to move further blue.

That 405 nm is worse than 532 nm, rather than better, follows from evaluating the absorption envelope at the temperature of the experiment (Sec. II C): the thermal phonon population supplies the anti-Stokes wing that keeps green excitation viable below the ZPL, and there is no counterpart to it on the blue flank, where the penalty is the intrinsic falloff of the Franck-Condon envelope.

The practical failure of green excitation is already visible in existing measurements. In Ref. [23] the PL spectra were recorded with 532 nm up to 41.4 GPa in one cell and with 457 nm from 55.2 GPa to 113.8 GPa in the other; the switch to blue at intermediate pressure was an operational necessity rather than a preference. The present analysis quantifies that necessity and extends it, since at 120 GPa even 457 nm leaves 10% of the sensitivity unclaimed.

The tolerance windows are 461.6–486.2 nm at the 5% level and 456.6–491.1 nm at the 10% level. The commercial 473 nm DPSS line lies 2.5 nm from the optimum and costs 0.2%; it is the recommended practical choice, and no custom source is required. The width of the window is itself worth noting: it is set by the breadth of the pressure-broadened sideband, which is a consequence of the increased electron-phonon coupling. The same physics that forces the move into the blue also makes

TABLE V. Anchor sensitivity of λ_{opt} at 120 GPa. Each parameter is displaced to the edges of its Monte Carlo range with all others held nominal. Entries below 0.01 nm are reported as exactly zero; they vanish by the cancellation of Eq. (12).

parameter	range	$\Delta\lambda_{\text{opt}} (-)$	$\Delta\lambda_{\text{opt}} (+)$
$\hbar\omega$ effective mode	$\pm 15\%$	6.88 nm	−6.66 nm
$\Delta E_{\text{ZPL}}(120)$	± 20 meV	3.65 nm	−3.60 nm
S_{abs} slope	$\pm 15\%$	2.73 nm	−2.70 nm
$dE_{\text{ZPL}}/dP _0$	± 0.25	0	0
σ_{ZPL}	$0.7\text{--}1.4\times$	0	0
a_{gs}	$\pm 50\%$	0	0
a_{es}	$\pm 50\%$	0	0
r_0	$\pm 30\%$	0	0
r_{bg}	$\pm 40\%$	0	0
w_0	$\pm 30\%$	0	0
C_{amb}	$\pm 50\%$	0	0
E_{ISC}	$\pm 30\%$	0	0

the choice forgiving once there.

D. Uncertainty of the optimum

For the legacy $\alpha = 0.95$ baseline, propagating the randomizable inputs of Table II through 400 Monte Carlo draws gives

$$\lambda_{\text{opt}} = 475.5^{+5.5}_{-5.4} \text{ nm} \quad (16\text{--}84\%), \quad (22)$$

with a standard deviation of about 5.4 nm. The band spans about half the 5% tolerance window, so at this level of input uncertainty the correct statement is not “475.5 nm” but “inside the 462–486 nm window, with 473 nm a safe commercial choice within this baseline.” This Monte Carlo band does not include the difference between absorption-model anchors or the two stress normalizations of Table III. The band includes a $\pm 15\%$ model-form range on the single effective phonon energy, which prevents the numerical precision of the grid search from being mistaken for physical precision. It is also, by construction, a propagation of *measurement* uncertainty: parameterizing the ZPL by $(\Delta E_{\text{ZPL}}(120), dE/dP|_0)$ instead of by (E_{max}, P_0) removes the parameterization slack that would otherwise let the 120 GPa shift wander far outside its measured uncertainty.

E. Which anchors control the answer

Table V is the most useful result of this section for planning purposes. At low intensity, λ_{opt} depends on the ZPL shift at the working pressure, the Huang-Rhys factor, and the effective-mode representation of the phonon sideband—and on nothing else. Every phenomenological charge-transfer constant, including the two ionization rates over ranges of $\pm 50\%$, and both constants of the contrast prefactor $C_0(P)$, leave λ_{opt} unchanged to numerical

precision. So does the initial slope of the ZPL shift, once the shift at 120 GPa is pinned: the answer depends on where the ZPL *is*, not on how it got there.

Two consequences follow. First, the low-intensity answer is considerably more trustworthy than the number of phenomenological knobs in the model would suggest: it inherits the uncertainty of quantities that Ref. [23] determined by DFT *and* measured directly, rather than the uncertainty of our parameterization. Expressed as derivatives,

$$\frac{\partial \lambda_{\text{opt}}}{\partial E_{\text{ZPL}}} \simeq -182 \text{ nm eV}^{-1}, \quad \frac{\partial \lambda_{\text{opt}}}{\partial S_{\text{abs}}} \simeq -11.9 \text{ nm}, \quad (23)$$

so a ZPL known to 20 meV fixes λ_{opt} to 3.6 nm, well inside the 5% tolerance window. The band of Sec. IV D is essentially that number combined with the $\pm 15\%$ ranges on the Huang-Rhys slope and the effective phonon energy, and it cannot be reduced except by measuring one of those quantities better.

Second, and less comfortably, the calibration programme of Sec. VII cannot improve the low-intensity answer at all. Its value lies elsewhere: in the absolute sensitivity, and in the intensity dependence, where the cancellation is expected to fail.

F. Pressure tracking

λ_{opt} is a property of the pressure, not of the technique. Figure 1(c) shows that it follows the ZPL and the side-band edge at $0.4\text{--}0.6 \text{ nm GPa}^{-1}$, moving from 485.6 nm at 100 GPa to 475.5 nm at 120 GPa; at both pressures it coincides with the maximum of $\lambda \sigma_{\text{abs}}$. Over the 97–172 GPa window spanned by the (La,Ce)H₉ family [8] the total excursion exceeds the 10% tolerance band, so the excitation wavelength has to be selected for the target pressure. A cell intended to cover a wide pressure range would benefit from two lines—488 nm below $\simeq 110$ GPa and 473 nm above—rather than from a single compromise.

G. The detection band moves too

The emission band shifts with the ZPL just as the absorption band does, and by more: the emission Huang-Rhys factor rises from 3.39 to 5.25 and the emission Debye-Waller factor falls from 4.9% to below 1% [23], so the emission maximum, $E_{\text{ZPL}} - S_{\text{em}} \hbar \omega$, moves from 719 nm at ambient pressure to 619 nm at 120 GPa. A passband chosen at ambient pressure to pass NV[−] while rejecting the laser and NV⁰—in practice 650–800 nm—therefore collects a shrinking fraction of the light: 82% at ambient, 47% at 50 GPa, 26% at 120 GPa. Since $\eta \propto R^{-1/2}$, that alone costs a factor 2.0 in sensitivity at 120 GPa—comparable to the entire gain from optimizing the excitation.

TABLE VI. The legacy baseline recommendation reverses below 73 GPa. Ratio of the sensitivity of the two lines at the same pressure and the same optical power; values below unity mean green is the better choice. The 50 GPa row is the pressure at which Ref. [37] compared blue with green and found no distinct advantage.

P (GPa)	λ_{opt} (nm)	$\eta(532 \text{ nm})/\eta(473 \text{ nm})$
20	556	0.30
30	543	0.35
50	522	0.56
73	503	1.01
100	486	1.85
120	476	2.52
150	468	3.55

The remedy is available only *because* the excitation has moved. With 532 nm excitation a cut-on below 650 nm is impractical; with 473 nm it is routine. Moving the cut-on to 600 nm raises the collected fraction at 120 GPa from 26% to 62%, a further factor 1.6 in η . The two choices compound: relative to the conventional pairing of 532 nm excitation with a 650–800 nm passband, blue excitation with a re-tuned passband is worth a factor $\simeq 4$ in sensitivity, i.e. a factor $\simeq 16$ in the acquisition time of a magnetic map.

Being a property of the emission, the collection efficiency is independent of excitation wavelength, so it multiplies η by a common factor and leaves λ_{opt} , the tolerance windows, and Table IV exactly unchanged. It matters for absolute sensitivity, and it matters for interpreting published count rates (Sec. VIII C).

V. PRESSURE AND INTENSITY BOUNDS OF THE LEGACY BASELINE

A wavelength quoted without its domain of validity is not a usable recommendation. The answer of Eq. (21) is bounded on two sides, and both bounds are part of that baseline result: one in pressure, one in intensity. Neither bound is recomputed here for Eq. (20).

A. In pressure: a threshold at 73 GPa

Run at lower pressure, the same calculation returns the opposite recommendation, and this is the one place where the model can be tested against a published *null* result. Table VI gives $\eta(532 \text{ nm})/\eta(473 \text{ nm})$ as a function of pressure. Below 73 GPa the ratio is smaller than unity and green is the better choice, by a factor 3 at 20 GPa. Blue overtakes green at approximately 73 GPa, leads by 20% near 80 GPa, by 50% at 87 GPa, and by a factor 2 only at 102 GPa. The threshold depends on which blue line is used: 457 nm crosses at 84 GPa and 450 nm at 89 GPa, because a line further from λ_{opt} has to wait for a larger shift of the envelope before it pays. Like λ_{opt}

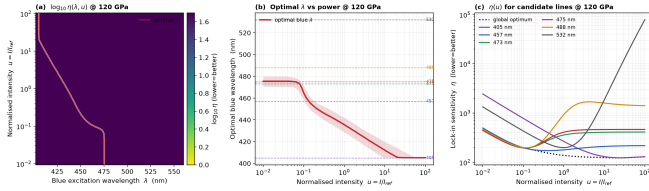


FIG. 3. Intensity-explicit model of Sec. IIF at 120 GPa: (a) $\log_{10} \eta(\lambda, u)$ with the optimal-wavelength ridge, (b) the ridge with its Monte Carlo band, (c) $\eta(u)$ for candidate laser lines. The dimensionless intensity u is defined so that $u = 1$ half-saturates the NV^- transition at λ_{opt} at this pressure; the linear photoluminescence measured at 100 GPa by Ref. [15] places existing experiments at $u \lesssim 0.3$.

itself, the crossover survives displacement of every phenomenological charge-transfer constant, moving by less than 2 GPa.

This reproduces an experiment that found nothing. Reference [37] compared 450 nm with 532 nm cwODMR on a [100] culet at 50 GPa and reported no distinct advantage for the blue line. The model places that measurement 39 GPa below its own 450 nm crossover, in a regime where it predicts green to be ahead. A positive reproduction can always be arranged by adjusting parameters; predicting where an effect is *absent* constrains the structure of the model rather than its constants, and among the observations available in the literature this is the one that does the most work.

It also settles the design of the experiment that would test the present analysis. A comparison of two excitation wavelengths at a single pressure measures a ratio that carries the relative calibration of the two laser powers, the relative transmission of anvil and optics at the two wavelengths, and the relative detection efficiency. Those systematics reach a factor of two without difficulty and are not separable from the predicted factor 2.5. All of them are, however, pressure independent, whereas the prediction is not: the same cell, the same two lines and the same optics, swept from 40–120 GPa, must show green ahead at the bottom, blue ahead at the top, and a crossing near 70 GPa. A multiplicative systematic can move a ratio, but it cannot create a crossing or displace one, since it does not depend on pressure. The sign change, not the ratio at one pressure, is the falsifiable measurement.

B. In intensity: the optimum runs blue with power

Whether λ_{opt} moves with laser power has direct consequences in a DAC, where the usable power is capped by sample heating rather than by the source. With the intensity unit fixed as in Sec. IIF, the low-intensity limit reproduces Eq. (21)— $\lambda_{\text{opt}}(u \rightarrow 0) = 475.5$ nm, verified as a regression test—and the question can be asked.

Direction. The naive expectation is that λ_{opt} should

move *red* with power, because two-photon ionization $\propto \sigma_{\text{abs}}^2 u^2$ penalizes excitation directly on the absorption peak. That expectation is wrong, and the reason is a sign. It rests on the NV^0 envelope lying red of the NV^- envelope; it does not. The NV^0 ZPL is 0.21 eV *above* the NV^- ZPL, so the recombination envelope lies to the *blue*, peaking near 439 nm at 120 GPa. Recombination is therefore fastest 35 nm blue of the NV^- peak, and both terms push λ_{opt} toward shorter wavelength. Reversing the sign of the NV^0 displacement in the model reverses the ridge, from 476–405 nm to 476–554 nm, confirming that this single input controls the direction.

Magnitude. At 120 GPa the ridge runs 475.5 nm ($u \rightarrow 0$) $\rightarrow$ 475.0 nm ($u = 0.06$) $\rightarrow$ 464.8 nm ($u = 0.1$) $\rightarrow$ 446.7 nm ($u = 0.3$), and saturates against the $\text{IP}(^3\text{A}_2)$ edge at 405.2 nm above $u \simeq 30$; it cannot pass that edge. Over 250 Monte Carlo draws of the intensity-dependent knobs the direction is unanimous: every draw places $\lambda_{\text{opt}}(u \geq 0.1)$ blue of its own $u \rightarrow 0$ value.

Where experiments actually sit. Reference [15] measured the photoluminescence of NV centers at 100 GPa as a function of excitation power and found it *linear*, concluding that a single-photon process dominates. In the present units that observation places the measurement at $u \lesssim 0.3$, since $R(u)$ departs from linearity by 10% at $u \simeq 0.28$. This is the operationally important statement: over $u < 0.06$ the optimum moves by less than 1 nm, but by $u = 0.1$ it has moved about 11 nm and left the 5% tolerance window, and by $u = 0.2$ it has left the 10% window. The intensity dependence is therefore not negligible—it becomes comparable to the tolerance band just below the onset of saturation, which is where one would naturally operate to maximize photon rate.

The cost of ignoring this is not small. If the recommendation of Sec. IV is followed literally—473 nm, selected in the $u \rightarrow 0$ limit—while the measurement is made at finite power, the penalty relative to the optimum *at that power* is 1.006 at $u = 0.1$, 1.22 at $u = 0.2$, and 1.51 at $u = 0.3$, where the optimum has moved to 447 nm. At the upper end of the range in which Ref. [15] places existing high-pressure measurements, that penalty is comparable to the factor 2.5 claimed here over green excitation: the wavelength error would consume most of the gain the wavelength choice was made to secure. The headline number is a statement about a regime, not about the technique. It is correct for $u \lesssim 0.1$; above that the recommendation moves into the blue with power, and a wavelength quoted without the power at which it applies is incomplete.

We do not replace 473 nm with a numerically different recommendation, because the magnitude of the shift depends on constants that are not yet calibrated. What follows instead is a condition and a measurement. The condition is $u \lesssim 0.1$. The measurement that converts it into an experimental power is a single saturation curve $R(I)$ at the target pressure: u is defined by a half-saturation condition, so the intensity at which $R(I)$ departs from linearity fixes the scale directly. That measurement is

TABLE VII. Classes of published observation reproduced by the frozen model. Four of these are the calibration anchors themselves (the ZPL slope and shift, the contrast decay, the stress anisotropy); the rest are independent. One observation is recorded as unreproduced.

class	source
ZPL shift, slope and saturation	[10, 23, 28]
405 nm ODMR absent at 17 GPa	[37]
405 nm ODMR appears near 100 GPa	[37]
NV ⁰ -dominated PL at low P under 405 nm	[37]
no blue advantage at 50 GPa (null)	[37]
532 nm cwODMR survives to 140 GPa	[15]
532 nm sensing at 150 GPa	[37]
integrated PL yield domes vs pressure	[23]
emission band blue-shifts with the ZPL	[23]
filtered confocal count rate falls with P	[15]
contrast 14% $\rightarrow$ 1–3% plateau	[15]
excitation stepped down with pressure	[14, 23]
ZPL shift per compressed volume, two geometries	[14]
micropillar contrast at 73, 103, 131 GPa	[14]
PL linear in power at 100 GPa	[15]

considerably cheaper than the multi-wavelength comparison and must precede it, since it determines which wavelengths the comparison ought to use. It is Stage 1 of Sec. VII.

The corresponding optimum in intensity is shallow. At 473 nm, $\eta(u)$ is minimized at $u^* \simeq 0.11$ and is only 30% worse at $u = 0.3$, so there is little to gain by pushing power in a DAC where the same power heats the sample.

What is and is not established. The *direction* of the shift and the existence of the 405.2 nm floor follow from the structure of the model—the relative placement of the two Franck-Condon envelopes and the position of the ionization edge—and both are anchored quantities. The *magnitude* is not: it depends on a'_{es} , on the saturation scales s_d and s_{d0} , and on our construction of the NV⁰ envelope, which borrows the NV⁻ Huang-Rhys factor and carries no zero-phonon line of its own. Those are exactly the constants the calibration of Sec. VII is designed to fix, and the intensity dependence—unlike the low-intensity optimum—is genuinely sensitive to them.

VI. VALIDATION AGAINST PUBLISHED OBSERVATIONS

Because the model contains phenomenological constants, its credibility rests on what it reproduces without them being adjusted. We therefore check it against every quantitative high-pressure observation we could extract from the literature, with the constants frozen at the values of Table II and the temperature set to that of each experiment. The suite comprises twenty-two published observations spanning 0–150 GPa and four excitation wavelengths; twenty-one are reproduced and one is recorded as unreproduced. Table VII summarizes the

classes; the complete suite, with sources and verdicts, is generated by the code of Appendix A.

Three of these deserve comment because they were not free.

The survival of green excitation to megabar pressure is the observation that forced the finite-temperature envelope of Sec. II C. With a $T = 0$ envelope the model predicts $\eta(532)/\eta_{\text{opt}} = 110$ at 140 GPa—green is dead—which contradicts Refs. [15, 37] outright. At 300 K the density is 2.5 at 120 GPa and below 6 at 150 GPa, and green survives as the measurements require. Since a single effective mode of 65 meV has $\bar{n} = 0.088$ at 300 K while the real phonon spectrum extends to much softer modes, the thermal wing computed here is a lower bound and the green penalty an upper one.

The 405 nm ODMR resonance that is absent at 17 GPa and present near 100 GPa follows from the ionization edge moving: at low pressure 405 nm lies far above IP(A_2), and drives the ensemble to NV⁰, whereas by 100 GPa the threshold has risen to meet it. No parameter was adjusted to place that transition.

The one observation not reproduced is the initial rise of the confocal count rate. Reference [15] reports a monotonic fall from ambient pressure, while the model—even after the collection factor of Sec. IV G is included, which moves the modelled peak down to 28 GPa and takes the 140 GPa rate to 3% of ambient—retains a rise below 30 GPa. We do not force it; the discrepancy is discussed in Sec. VIII C.

VII. EXPERIMENTAL PROGRAMME

The analysis above specifies two absorption-model hypotheses and leaves their anisotropic comparison open. For the legacy low-intensity baseline, under quasi-hydrostatic stress, at room temperature, in the unsaturated regime, and at equal incident optical power, the model centre value is 475.5 nm and the commercial 473 nm line is within 0.2% of it. The hydrostatically anchored correction instead gives the α -dependent candidates of Table III. A measured stress state and an excitation sweep should therefore precede a final wavelength choice. What is not fixed is the *experimental* power that “unsaturated” corresponds to, and the magnitude of the shift beyond it. The programme below is staged so that each stage is useful on its own and determines whether the next is necessary.

A. Cell and sensor

The measurements are planned in a DAC with a near-surface, high-density NV layer implanted in the culet of the lower anvil. Micropillars machined on a (100)-oriented culet [14] reduce the non-hydrostatic stress on the NV layer to $\alpha \simeq 0.95$ and keep all four NV orientations usable, which provides a near-hydrostatic validation

geometry. The hydrostatic absorption anchor still requires the conditional correction, or an experimental test of it, described in Sec. IV A. A (111)-oriented anvil [18] instead selects a single NV orientation aligned with the compression axis; it reaches higher pressure but the ZPL splitting and the modified intersystem crossing [24] would require the anchors to be re-derived. Silver-coating the tip of the NV-implanted anvil suppresses interference from the opposing anvil and improves collection efficiency [23]. Argon is the pressure-transmitting medium [38]; pressure is read from $\text{SrB}_4\text{O}_7\text{:Sm}^{2+}$ at low pressure and from the first-order diamond Raman edge above [39]. Because the recommendation is a blue one, the NV layer must be implanted into a low-nitrogen anvil rather than bulk type Ib (Sec. VIII D).

B. Stage 1: fix the intensity scale

A single-wavelength power sweep at the legacy comparison line 473 nm and the target pressure, recording at each incident power the power density at the sample, the PL rate $R(I)$, the ODMR contrast $C(I)$, the linewidth $\Delta\nu(I)$, and an indicator of sample temperature. From these, $\eta(I) \propto \Delta\nu(I)/[C(I)\sqrt{R(I)}]$.

Stage 1 determines the powers at which R departs from linearity by 5% and 10%, the half-saturation power $I_{1/2}$, the powers at which contrast degradation and power broadening begin, the sensitivity-optimal power at 473 nm, and the maximum power permitted by heating. It thereby converts the condition $u \lesssim 0.1$ of Sec. V B into an experimental number. This is the cheapest measurement in the programme and everything else depends on it: if $R \propto I$ with C and $\Delta\nu$ constant across the intended operating range, the low-intensity comparison at 473 nm can be made in the unsaturated regime; the wavelength choice still requires Stage 2 to distinguish the absorption anchors and their stress dependence. Saturation, contrast loss, or broadening within that range additionally requires a finite-power comparison.

C. Stage 2: the multi-wavelength comparison

The wavelength ordering must be measured both to test the stress correction and, if Stage 1 finds finite u , to quantify the intensity shift. We retain 457, 473 and 488 nm, with 405 nm added if available to probe the ground-state ionization edge at high power. A tunable sweep or an additional line near 441–446 nm is needed to test the hydrostatic and micropillar corrected optima; the legacy three-line comparison alone does not bracket them. The conditions that must be held fixed are the pressure, the sample position, the focusing, the microwave drive, and the detection bandwidth; the incident power must be equalized *at the sample*, calibrated rather than read from the laser head, and the spot size, anvil and objective transmission, and detection efficiency

recorded per wavelength. Each line is swept over the same power-density range, from the linear regime established in Stage 1 into saturation, and η computed from the same three observables.

The analysis extracts $\lambda_{\text{best}}(I) = \arg \min_{\lambda} \eta(\lambda, I)$ over the measured lines. Three outcomes are anticipated. If 473 nm stays within 5% of the best line across the operating range, the recommendation is frozen as power independent in practice. If 457 nm overtakes it at high power by a statistically significant margin, the switching power is reported and the recommendation is split by regime—which is what Sec. V B predicts. If the measurement uncertainty exceeds the differences between lines, the result is reported as an equivalence band rather than a single wavelength. With only three commercial lines the claim must be “best of the lines measured,” not “the continuous optimum”; a continuous claim requires additional wavelengths for local interpolation.

D. Stage 3: calibrating the charge-conversion model

Claiming a *mechanism* for the finite-power shift—rather than reporting the shift—requires separating optical saturation from charge conversion, which $R(I)$ alone cannot do: a rollover in PL may be excitation saturation, NV^- depletion, or heating. Stage 3 therefore adds NV^-/NV^0 spectral separation at each wavelength and power, giving $f_-(\lambda, I)$ directly; time-resolved ionization and recombination rates where feasible; a two-beam (I_{405}, I_{457}) sweep, each varied over at least two decades; and a measurement of laser-induced local heating. The two-beam design is deliberate: 405 nm is the only accessible wavelength at which the ground-state ionization term of Eq. (10) is open at 120 GPa, so it isolates a_{gs} , while 457 nm sits well inside the sideband where only the σ_{abs} -proportional channels operate. Dual-beam schemes of this kind have been used to separate the charge-state channels at ambient pressure [40].

The fit targets $\{a_{\text{gs}}, a'_{\text{es}}, r_0, r_{\text{bg}}, s_d, s_{d0}, w_0\}$ by least squares with a Markov-chain posterior. Before the measurement, an identifiability check on synthetic data is required: the two-photon coefficient a'_{es} and the saturation scale enter the low-power expansion in the same combination, and a profile-likelihood or Fisher-information analysis should establish whether the planned sweep separates them. If it does not, the remedy is to add single-colour sweeps at 457 nm and 488 nm, whose different σ_{abs} break the degeneracy at fixed u .

E. Controls

Independent of stage, the following controls are required: a power sweep at ambient or low pressure through the same optics; per-wavelength calibration of the power delivered to the sample and of the spot size; measure-

ments at fixed microwave power and with the microwave re-optimized at each optical power; ascending and descending sweeps to detect hysteresis and irreversible change; and a long-illumination run to bound drift, NV bleaching, and sample modification.

F. What the programme tests

Within the legacy baseline, λ_{opt} at low intensity is insensitive to every parameter this calibration determines (Table V), the calibration is not a check on the 475.5 nm result. The added stress-resolved excitation sweep directly tests that baseline against the corrected Ho-anchor hypothesis. The remaining calibration tests four other things. Stage 1 fixes the regime, converting $u \lesssim 0.1$ into milliwatts. Stage 2 tests the predicted direction of the finite-power shift. Stage 3 fixes the absolute sensitivity scale, converting η into nT/ $\sqrt{\text{Hz}}$, and provides the first direct measurement of f_- under compression: the model predicts $f_- \simeq 0.58$ across the blue window, wavelength independent to 0.2%, and a measured wavelength dependence of f_- would falsify the structure of Eqs. (10)–(11) directly.

The quantitative predictions available for comparison are collected in Table VIII. The last row is the one to prioritize: as Sec. V A argues, the pressure at which the ordering of green and blue reverses is the only prediction here that a pressure-independent systematic cannot fake.

VIII. DISCUSSION

A. Scope and tests of the stress correction

Section IV A makes the stress dependence explicit before presenting the legacy baseline. The two calculations share a phenomenological description of the differential stress shift but have different absorption anchors: the single-mode envelope and the published hydrostatic Ho kernel. The optical predictions concern relative sensitivity at low intensity; absolute sensitivity still requires Stage 3 of Sec. VII.

In the axial-stress convention of Eq. (19), Ref. [14] measured α directly from the ODMR field response: $\alpha = 0.56$ for a standard flat culet, $\alpha = 0.95$ for a FIB-machined micropillar. The same work measured the ZPL shift per unit compressed volume in both geometries— $-769 \text{ meV}/(\text{cm}^3 \text{ mol}^{-1})$ on the micropillar against -434 on the flat culet—so at equal compression the deviatorically stressed culet shifts the ZPL by only 0.56 of the quasi-hydrostatic value. Deviatoric stress red-shifts the 3E manifold and thereby opposes the hydrostatic blue shift, as Davies and Hamer [41] established at low stress. The differential calculation propagates two readings of this ratio rather than treating the stress coefficient as uniquely known. Its limits and rigid-shape assumption are stated in Fig. 2.

For auditability, Table IX retains the earlier single-mode calibration, which held the micropillar value unchanged by construction. Its flat-culet result, 510 nm with green outperforming 473 nm, is a legacy model result. It is distinct from the 478.9–493.4 nm conditional range at $\alpha = 0.56$ in Table III; the latter uses the Ho hydrostatic anchor and only a differential envelope shift. Neither is an experimentally validated flat-culet optimum.

The optical correction alone does not predict whether a usable ODMR signal exists. Deviatoric stress also disrupts the ODMR contrast: Ref. [14] reports that a flat culet loses practical magnetic sensitivity above 40–50 GPa, with the contrast first vanishing near 40 GPa and then inverting sign, while the micropillar retains negative contrast of 5%, 3% and 1.5% at 73, 103 and 131 GPa. The *ab initio* calculation of Ref. [24] explains why: symmetry-breaking stress lifts the 3E orbital degeneracy and opens intersystem-crossing channels that are absent under hydrostatic compression, and around 60 GPa in a (100)-cut anvil the lower-branch crossing becomes spin-selective toward the dark sublevel, inverting the sign of the ground-state polarization—and hence of the contrast—rather than merely suppressing it. No sensitivity has been computed for that inverted-contrast regime, so Table IX does not use it; “not an operating point” here means “not modelled,” not “physically dead.”

A quasi-hydrostatic geometry provides a useful validation setting, but the Ho absorption kernel is anchored at $\alpha = 1$, not at the measured micropillar $\alpha = 0.95$. The candidate correction at the latter is already 444.7–446.1 nm. A further route—a (111)-cut culet, in which the 3E of the axial NV group red-shifts and the intersystem crossing is preserved [37]—is expected to behave similarly, and Ref. [24] finds *ab initio* that near-uniaxial stress along [111] is not merely tolerable but genuinely optimal for the contrast of that NV subensemble. Neither a measurement of α nor of the ZPL shift exists yet for that geometry in the convention of Table I, so we do not extrapolate Table IX to it.

B. Is the contrast wavelength independent?

The optical wavelength comparisons in Sec. IV assume that C does not depend on the excitation wavelength, so that minimizing η is maximizing $\lambda\sigma_{\text{abs}}$. The assumption has a specific way to fail. Under stress the 3E orbital doublet splits, with an orbital-strain coupling of $850 \pm 130 \text{ THz}$ per unit strain [42]; if the two branches had different intersystem-crossing rates, and if the excitation wavelength selected between them, then $C = C(\lambda)$ and the optimum would move. This is the mechanism suggested in Ref. [37] as a way to measure ISC rates from different orbital states, and it is worth taking seriously precisely because it would be useful if true.

It does not survive quantitatively, for two independent

TABLE VIII. Legacy single-mode baseline predictions to be compared against measurement, distinct from Table III. All entries are at 120 GPa, room temperature, $\alpha = 0.95$ and $u \lesssim 0.1$ unless stated.

quantity	prediction
λ_{opt}	475.5 nm ± 5.5
$\eta(473)/\eta(\lambda_{\text{opt}})$	1.002
$\eta(457)/\eta(\lambda_{\text{opt}})$	1.11
$\eta(532)/\eta(\lambda_{\text{opt}})$	2.53
$\eta(405)/\eta(\lambda_{\text{opt}})$	3.76
5% tolerance window	461.6–486.2 nm
f_- across that window	0.582, flat to 0.2%
collected fraction, 650–800 nm	26%
λ_{opt} at 100 GPa	485.6 nm
$\lambda_{\text{opt}}(u)$ direction	moves blue; floor at 405.2 nm
green/blue crossover (473 nm)	73 GPa

TABLE IX. Legacy single-mode calibration retained for comparison, not the hydrostatically anchored correction of Table III. Dependence of its optimum on the stress anisotropy $\alpha = \sigma_{xx}/\sigma_{zz} = \sigma_{yy}/\sigma_{zz}$ at the culet, at 120 GPa and room temperature, calibrated to the two geometries measured in Ref. [14]. The $\alpha = 0.56$ row is not a straightforward operating point: a flat culet drives the ODMR contrast through zero near 40–50 GPa, after which it recovers with inverted sign [14, 24].

geometry	α	$\Delta E_{\text{ZPL}}(120)$	λ_{opt}	$\eta(532)/\eta(473)$
ideal hydrostatic	1.00	422 meV	471 nm	3.06
micropillar [14, 23]	0.95	400 meV	476 nm	2.52
intermediate	0.80	333 meV	488 nm	1.63
standard flat culet [14]	0.56	226 meV	510 nm	0.82

reasons.

The splitting is small compared with $k_B T$. In the micropillar geometry the deviatoric stress at 120 GPa is $(1 - \alpha)P = 6$ GPa, giving a strain of $5\text{--}10 \times 10^{-3}$ depending on whether one uses Young’s modulus or C_{44} , and hence a branch splitting $\delta = 19 - -36$ meV against $k_B T = 26$ meV. The branches are thermally mixed, not resolved. For comparison, the Franck-Condon envelope is 140 meV wide, so the two branch envelopes overlap almost completely and the fraction of absorption going into either branch varies by only 3–6 percentage points across the whole 5% tolerance window.

The population cannot remember which branch absorbed the photon. Downward orbital relaxation in 3E is spontaneous phonon emission: it requires no thermal activation and proceeds on the picosecond scale whatever δ is. That is the same relaxation that motionally averages the excited-state zero-field splitting to 1.42 GHz at room temperature [43]. The intersystem crossing, at $\Gamma_{\text{ISC}} \sim 3 \times 10^7 \text{ s}^{-1}$, is four orders of magnitude slower. The branch populations therefore reach thermal equilibrium—a distribution fixed by δ and T alone—roughly 10^4 relaxation times before a single ISC event. The contrast is set by the relaxed 3E , for exactly the reason the emission spectrum is (Kasha’s rule). The same argument disposes of the vibrational analogue: a vibrationally hot 3E has a larger $^3E\text{--}^1A_1$ gap and hence a different ISC rate, but vibrational relaxation is also picosecond.

Because the first argument alone would be

uncomfortable— $\delta \simeq k_B T$ is not a large margin—we also bound the worst case, granting the failure mechanism everything it needs: no orbital relaxation at all, so the branch populated by absorption is the branch that crosses. Writing $C(\lambda) = r(\lambda)C_A + [1 - r(\lambda)]C_B$ with r the branch-selection ratio computed from the two displaced envelopes, and weighting only C (which enters η linearly, unlike $\sqrt{R}$, so this maximizes the effect), a contrast ratio C_B/C_A anywhere between 0 and 10 moves λ_{opt} by at most 3.8 nm at $\delta = 19$ meV and 7.3 nm at $\delta = 36$ meV—inside the 5% window in every case. Even with one branch completely dark, δ must exceed 63 meV before λ_{opt} leaves that window, a margin of 1.7–3.3 over the micropillar value. The threshold is comfortably exceeded in a flat culet ($\delta \simeq 170 - -320$ meV), which is one more reason that geometry is unusable, and it is another instance of the pattern of Sec. VIII A: what makes the answer robust is the same quasi-hydrostatic condition that makes the measurement possible at all.

The direct test is cheap and does not require new hardware: measure the ODMR *contrast*—not the signal-to-noise ratio, which convolves contrast with photon rate—at two excitation wavelengths at the same pressure and the same optical pumping rate. Reference [37] already recorded cwODMR spectra at 532 and 450 nm on a (100) culet near 50 GPa; extracting the two contrasts from that data would settle the question at moderate pressure immediately.

C. Two measurements of the pressure dependence of the photoluminescence

Two published measurements of how NV fluorescence under 532 nm excitation evolves with pressure appear to disagree, and the disagreement is instructive. Reference [23] reports the spectrally integrated PL intensity at constant laser power and integration time and finds a broad *dome*—a rise to a maximum at intermediate pressure, then a decline—which it attributes to the absorption cross section sweeping past the fixed laser line, noting that the radiative quantum efficiency is stable to 120 GPa. Reference [15] reports a confocal count rate and finds a dramatic fall over the first 50 GPa followed by a slow decline.

These are different observables. The model computes the first: $R \propto f_- \lambda \sigma_{\text{abs}}$ is the rate of absorbed photons, and it reproduces the dome. A count rate additionally carries the collection efficiency of Sec. IV G, which falls by a factor 3.2 from ambient to 140 GPa. Including it moves the modelled peak down to 28 GPa and takes the 140 GPa count rate to 3% of ambient. The filter thus accounts for most of the high-pressure decline but not for the initial rise, which the model retains and Ref. [15] does not observe.

We do not force that residual. It is a factor of roughly two, confined below 30 GPa, and the quantity it concerns has been measured directly and found to rise [23]. The discrepancy is therefore between two experiments rather than between the model and experiment, and the most likely origin is instrumental: Ref. [15] tracks a micron-sized diamond particle in a compressing pressure medium, where focus, collection solid angle and medium transparency all drift, whereas Ref. [23] works with an NV layer fixed in an anvil micropillar. The discriminating measurement is straightforward—spectrally integrated PL yield at constant power over 0–50 GPa under 532 nm excitation—and it costs one pressure ramp in the cell of Sec. VII A.

D. Delivering blue light through the anvil

A practical obstacle stands between the recommendation and its use. Type Ib diamond, the standard host for high-density NV ensembles, absorbs strongly below 500 nm through its substitutional-nitrogen continuum. In Ref. [37] this forced the excitation to be delivered through the opposite, untreated anvil, which restricted the measurement to the NV centers in the sample-chamber region. A 473 nm recommendation that cannot be delivered through the sensing anvil is not a recommendation.

The route taken by Refs. [14, 23] removes the obstacle: the NV layer is created by nitrogen focused-ion-beam implantation into a low-nitrogen anvil followed by annealing, so the bulk of the anvil never carries the type Ib nitrogen continuum and the excitation is delivered through

the sensing anvil itself. Consistently, that work stepped the excitation down through 532, 488, 457 and 405 nm as pressure increased, explicitly to follow the absorption blue shift—the practice this paper quantifies. Any implementation of the present recommendation must use implantation into a low-nitrogen anvil rather than a bulk type Ib anvil; we note in passing that our model places 457–473 nm well ahead of 405 nm at 100–130 GPa, so the final step of that sequence appears to overshoot, which is directly testable against the existing data.

E. Limitations

Several approximations deserve to be named. The single effective phonon mode of Eq. (7) reproduces the envelope but not the residual structure of the first replica; at 120 GPa this is benign, because Ref. [23] reports that the structure has washed out, but it would be a poor approximation below 50 GPa. It is also the largest single entry in Table V, which is why $\hbar\omega$ is randomized rather than held fixed. The dynamical Jahn-Teller contribution to absorption is folded into the effective S_{abs} rather than treated explicitly through the squared vibronic coupling constants K^2 , which rise from 0.58 to 0.97 under compression; since λ_{opt} depends on the envelope maximum, and K^2 affects primarily the width and the fine structure, this is expected to shift λ_{opt} less than the S_{abs} uncertainty already tabulated, but it has not been verified.

The lineshapes of Ref. [23] were obtained at 30 K and $\simeq 90$ K, whereas the envelope here is evaluated at the temperature of the intended experiment. This is a strength rather than an extrapolation— T enters Eq. (7) explicitly—but the single-mode restriction means the thermal wing is a lower bound, as noted in Sec. VI.

Even in the micropillar geometry with $\alpha \simeq 0.95$ the 3E degeneracy is lifted and the ZPL splits, with only the lower branch reproduced by the calculation of Ref. [23]; we use a single ZPL. Given $\partial\lambda_{\text{opt}}/\partial E_{\text{ZPL}} \simeq -182 \text{ nm eV}^{-1}$, a splitting of 20–40 meV corresponds to 4–7 nm of ambiguity in λ_{opt} —a reason to prefer a source near the center of the tolerance window rather than at its edge.

The contrast prefactor $C_0(P)$ is a single decaying exponential calibrated to the hydrostatic, randomly oriented case [15]; it is wavelength independent and therefore cannot move λ_{opt} , but it should not be read as a prediction of contrast for a specific culet cut. The *ab initio* intersystem-crossing rate of Ref. [24] is not monotonic even under pure hydrostatic stress—it peaks near 30 GPa before falling—and under deviatoric stress it can invert the sign of the contrast outright; $C_0(P)$ captures none of this structure, only the net decay observed at 120 GPa by Ref. [15].

Finally, sample heating is not modelled. At fixed low power the neglect is harmless, but the moment intensity dependence is addressed the absorbed power at the culet

shifts the local ZPL and hence λ_{opt} itself, a feedback the present model does not contain. This is why Stage 1 of Sec. VII records a temperature indicator alongside the optical observables.

F. Relation to existing practice

The guideline stated in Ref. [23]—tune the excitation to follow the ZPL shift—is confirmed and made quantitative. What the present analysis adds is the specific target, the tolerance, the identification of 405 nm as a boundary rather than an option, the observation that within the window the answer is controlled entirely by measured optical quantities and not by charge-transfer phenomenology, and the two bounds of Sec. V. Reference [14] already stepped its excitation down through 532, 488, 457 and 405 nm with rising pressure; the present model reproduces the direction and the ordering of that sequence. A 473 nm DPSS source and a detection cut-on moved to 600 nm are the practical legacy baseline. A final source choice requires the stress-dependent comparison of Sec. IV A, which predicts shorter wavelengths near the hydrostatic limit when anchored to the Ho kernel.

IX. SUMMARY AND OUTLOOK

The pressure-induced reorganization of NV photo-physics at 120 GPa—a ZPL blue shift beyond 0.4 eV, a 50% increase of the absorption Huang-Rhys factor, and a rise of the ground-state photoionization threshold to 3.06 eV—places the legacy single-mode ODMR optimum in a window centered at 475.5 nm, for $\alpha = 0.95$. Both conventional choices lie outside it, for opposite reasons: 532 nm because the absorption onset has moved past it, 405 nm because it lies on the far blue flank with the ionization edge immediately beyond.

Within the window the steady-state charge state is nearly wavelength independent, because excited-state ionization and recombination share the same σ_{abs} dependence and cancel from f_- . At equal optical power the optimum is therefore the maximum of $\lambda\sigma_{\text{abs}}$, and—as the anchor analysis shows—it is determined entirely by the ZPL position and the electron-phonon coupling, with no sensitivity to any of the phenomenological charge-transfer constants. Within that legacy baseline, a 473 nm source recovers essentially the full optimum, buying back a factor 2.5 in sensitivity relative to green excitation, equivalently a factor $\simeq 6$ in the acquisition time of a magnetic map, with a further factor 1.6 available because blue excitation is what permits the detection band to be moved with the emission.

The hydrostatically anchored alternative changes the design target: Eq. (20) fixes 440.65 nm at $\alpha = 1$ and predicts 444.7–446.1 nm at $\alpha = 0.95$, 466.1–475.4 nm at $\alpha = 0.70$, and 478.9–493.4 nm at $\alpha = 0.56$, at fixed axial stress 120 GPa. The ranges span two model normaliza-

tions and are unvalidated. The stress coefficient and the anisotropic absorption shape must be measured before turning these candidate wavelengths into a source recommendation.

The legacy answer also has pressure and intensity bounds. It is bounded in pressure: below 73 GPa green is the better choice, which is why a comparison at 50 GPa found no advantage for blue, and the crossing between the two regimes is the prediction that a single-pressure measurement cannot fake. It is bounded in intensity: the optimum moves into the blue as the transition saturates, so 473 nm is the baseline answer for $u \lesssim 0.1$. These bounds have not been recalculated for the hydrostatically anchored stress correction.

The stress state and the low-intensity excitation spectrum should be measured alongside the following programme. A saturation curve at the target pressure fixes the intensity scale and therefore decides which of the two power regimes an experiment is in; everything else depends on it. A multi-wavelength comparison at that pressure tests the predicted direction of the finite-power shift. And a pressure sweep through the green/blue crossing tests the prediction that is least exposed to systematic error, precisely because the quantity being predicted—the pressure at which the ordering reverses—does not depend on any of the multiplicative factors that a wavelength comparison must otherwise control.

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[To be completed.]

Appendix A: Reproducing the numbers

The legacy baseline figures and quoted values are generated by the code in `code/`; the differential stress correction is reproduced separately:

- `calc_alpha_corrected_optimum.py` — the Ho-anchored differential stress correction and both normalizations of Table III.
- `nv_model.py` — anchored optical inputs, σ_{abs} , the emission envelope and collection factor, f_- , and η at fixed power; Monte Carlo helper.
- `nv_model_power.py` — the intensity-explicit extension of Sec. II F.
- `fig1_green_blue_mix.py` — green/blue/mixed excitation versus pressure.
- `fig2_blue_wavelength_sweep.py` — Fig. 1.
- `fig3_power_sweep.py` — Fig. 3.
- `repro_literature.py` — the validation suite of Sec. VI: every published observation the model is checked against, with its source and its verdict.

- `analysis_C_lambda.py` — the orbital-branch analysis of Sec. VIII B.

- `tests/test_freeze.py` — the regression suite that pins the frozen numbers, including bit-exact backward compatibility of the $T = 0$ limit.

```
cd code
# alpha dependence, hydrostatic anchor at 120 GPa
python calc_alpha_corrected_optimum.py
# green/blue crossover
python figures/fig1_green_blue_mix.py
# 120 GPa (vs 100, 140)
python figures/fig2_blue_wavelength_sweep.py
python figures/fig2_blue_wavelength_sweep.py 100 75 125
# power dependence
python figures/fig3_power_sweep.py
python repro_literature.py           # the validation suite
python analysis_C_lambda.py         # the C(lambda) analysis
python -m pytest tests/ -q          # the freeze tests
```

Legacy baseline regression values pinned by the tests: $\lambda_{\text{opt}} = 485.6$ nm at 100 GPa and 475.5 nm at 120 GPa; the 473 nm, 457 nm and 450 nm crossovers against 532 nm

at 73 GPa, 84 GPa and 89 GPa; $\eta(457 \text{ nm})/\eta(\lambda_{\text{opt}}) = 1.11$ at 120 GPa; half-saturation at $u = 1$ by definition; and $\lambda_{\text{opt}}(u \rightarrow 0) = \lambda_{\text{opt}}$ of the fixed-power model.

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